

Angular Momentum & 3D Schrodinger Eq

Just how momentum operator $e^{\frac{i\hat{p}a}{\hbar}}$ generates translations.
Angular momentum operator generates Rotations.

Recall: $\hat{p} = (\hat{p}_x, \hat{p}_y, \hat{p}_z) = \frac{\hbar}{i} \nabla = \frac{\hbar}{i} \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$

In 3D:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(r) + V(r) \psi(r) = E \psi(r)$$

$$r \rightarrow r, \theta, \phi$$

$$r \rightarrow r$$

For central Potential
if not then $V(r)$ instead

In spherical coord:

$$\nabla^2 \psi = (\nabla \cdot \nabla) \psi = \frac{1}{r^2} \frac{\partial^2}{\partial r^2} (r^2 \psi) + \frac{1}{r^2} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) \psi$$

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} = -\frac{L^2}{\hbar^2}$$

$$L^2 = \hat{L}_x \hat{L}_x + \hat{L}_y \hat{L}_y + \hat{L}_z \hat{L}_z$$

Thus,

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V(r) \psi = E \psi$$

$\Rightarrow$ The Operator $\hat{L}$

$$L = (L_x, L_y, L_z) \equiv r \times p$$

So

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$$

A) Hermitian : $(\hat{L}_x)^\dagger = \hat{L}_x$

B) Commutator : $[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$: Are cyclic

$\hat{L}$: Orbital Angular Momentum

$\hat{S}$: Spin Angular Momentum.

* The (x, y, z) $\hat{L}$ states can't be simultaneous eigenstates

$\hookrightarrow$ Following up

$$\text{Casimir Operator : } \hat{L}^2$$

$$\hookrightarrow [\hat{L}_x, \hat{L}^2] = 0$$

$\Rightarrow$ In Spherical Coord :

$$\hat{L}_z = \hat{x} \hat{p}_y - \hat{y} \hat{p}_x$$

$$= \frac{\hbar}{i} \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned}$$

$$\frac{\partial}{\partial \phi} = \frac{\partial x}{\partial \phi} \frac{\partial}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial}{\partial y} + \cancel{\frac{\partial z}{\partial \phi} \frac{\partial}{\partial z}}$$

$$\frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi \quad ; \quad \frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi$$

$$= -y \quad \quad \quad = x$$

$$\text{So, } \frac{\partial}{\partial \phi} = x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x}$$

$$\Rightarrow \boxed{\hat{L}_z = \frac{\hbar}{i} \frac{\partial}{\partial \phi}} \quad : \text{Confirms } \hat{L} \text{ generates Rotations.}$$

$$\Rightarrow \boxed{-\frac{\hat{L}^2}{\hbar^2} = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2}}$$

$\Rightarrow$ Eigenstates of $\hat{L}$:

Finding simultaneous eigenstate of $\hat{L}^2$ & $\hat{L}_z$

$$\hat{L}_z \psi_{lm} = \hbar m \psi_{lm} \quad m \in \mathbb{R} \quad \text{--- (1)}$$

$$\hat{L}^2 \psi_{lm} = \hbar^2 l(l+1) \psi_{lm} \quad l \in \mathbb{R} \quad \text{--- (2)}$$

$$x \quad E_0 \quad 1 \quad 1 \quad \hat{L}^2 \quad 1 \quad 1 \quad 2 \quad 0 \quad 1 \quad 0 \quad 0 \quad (20 \quad \hat{L}^2 \quad 10) \geq 0$$

* Eigenvalues of L are always positive: $(\Psi, L\Psi) \geq 0$

Using $\hat{L}_z = \frac{\hbar}{i} \frac{\partial}{\partial \phi}$

① $\hookrightarrow \frac{\hbar}{i} \frac{\partial \psi_{lm}}{\partial \phi} = \hbar m \psi_{lm} \rightarrow \frac{\partial \psi_{lm}}{\partial \phi} = i m \psi_{lm}$

Solving $\rightarrow \boxed{\psi_{lm}(\theta, \phi) = e^{im\phi} P_l^m(\theta)}$

For ψ_{lm} to be defined as a func. of angles only:

$\hookrightarrow \psi_{lm}(\theta, \phi + 2\pi) = \psi_{lm}(\theta, \phi)$

$e^{im(\phi+2\pi)} = e^{im\phi} \rightarrow e^{2\pi im} = 1$

$\boxed{m \in \mathbb{Z}}$

② $\left(\sin\theta \frac{\partial}{\partial \theta} \sin\theta \frac{\partial}{\partial \theta} + \frac{\partial^2}{\partial \phi^2} \right) \psi_{lm} = -l(l+1) \sin^2\theta \psi_{lm}$

now put $\psi_{lm} = e^{im\phi} P_l^m(\theta)$

Thus,

$$\frac{d}{dn} \left[(1-n^2) \frac{dP_l^m}{dn} \right] + \left[l(l+1) - \frac{m^2}{1-n^2} \right] P_l^m(n) = 0$$

By putting $m=0$ we learn that:

$\hookrightarrow \boxed{l = 0, 1, 2, 3, \dots}$: Quantization

$$P_m^l = \frac{1}{2^l l!} \cdot \left(\frac{d}{dn} \right)^l (n^2 - 1)^l : P_0 = 1, P_1 = n, P_2 = \frac{1}{2}(3n^2 - 1)$$

comes from $\boxed{-l \leq m \leq l}$

$$P_l^m = (1 - n^2)^{|m|/2} \left(\frac{d}{dn} \right)^{|m|} P_l(n)$$

Thus, ψ_{lm} eigenfunctions (normalised) are called Spherical Harmonics $Y_{lm}(\theta, \phi)$ where,

$$Y_{lm}(\theta, \phi) = \underbrace{\sqrt{\frac{2l+1}{4\pi} \cdot \frac{(l-m)!}{(l+m)!}}} \cdot (-1)^m e^{im\phi} P_l^m(\cos\theta)$$

normalisation

For $m < 0$

$$\hookrightarrow Y_{lm}(\theta, \phi) = (-1)^m \cdot [Y_{l,-m}(\theta, \phi)]^*$$

Hence,

$$\begin{aligned} \hat{L}_z Y_{lm} &= \hbar m Y_{lm} \\ \hat{L}^2 Y_{lm} &= \hbar^2 l(l+1) Y_{lm} \end{aligned}$$

Radial Wave Equation

$$\Psi(r, \theta, \phi) = R_{El}(r) \cdot Y_{lm}(\theta, \phi)$$

insert this into SE:

$$\hbar^2 \nabla^2 \Psi = E \Psi \quad \left(\nabla^2 = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right) + \frac{1}{r^2} \nabla_{\theta\phi}^2 \right)$$

$$\boxed{-\frac{\hbar^2}{2m} \frac{d}{dr^2} (u_{\ell}) + \left(V(r) + \frac{\hbar^2 \ell(\ell+1)}{2mr^2} \right) u_{\ell} = E u_{\ell}}$$

$$V_{\text{eff}}(r)$$

$$\boxed{u_{\ell} \equiv r R_{\ell}}$$

$$V_{\text{eff}}(r) = V(r) + \underbrace{\frac{\hbar^2 \ell(\ell+1)}{2mr^2}}_{\text{centrifugal term}}$$

$$\Rightarrow \psi(r, \theta, \phi) = R_{\ell}(\gamma) \cdot Y_{\ell m}(\theta, \phi) = \frac{u_{\ell}(\gamma)}{r} Y_{\ell m}(\theta, \phi)$$

$$\bullet \int_0^{\infty} dr |u_{\ell}|^2 = 1$$

$\Rightarrow$ Behaviour at $r \rightarrow 0$

$\Rightarrow$ When the centrifugal dominates at $r \rightarrow 0$

$$-\frac{\hbar^2}{2m} \frac{d^2 u_{\ell}}{dr^2} + \frac{\hbar^2 \ell(\ell+1)}{2mr^2} u_{\ell} = 0 \quad \text{as } r \rightarrow 0$$

$$\frac{d^2 u_{\ell}}{dr^2} = \frac{\ell(\ell+1)}{r^2} u_{\ell}$$

$\hookrightarrow$ Solution: $u_{\ell} = r^s$

where

$$s(s-1) = \ell(\ell+1) \Rightarrow \underline{s = -\ell}$$

At $r=0$

$$\hookrightarrow u_{\ell} \sim r^{\ell+1} \quad \text{or} \quad u_{\ell} \sim \frac{1}{r^{\ell}}$$



For all $\ell \geq 0$

$$\boxed{U_{El} \sim c r^{l+1}} \quad \text{as } r \rightarrow 0$$

hope: U_{El} vanishes at $r=0$

$$\boxed{R_{El} \sim c r^l}$$

Constant non-zero wave-func. at origin only for $l=0$.
For $l \neq 0$, the angular momentum barrier prevents the particle from reaching the origin

$\Rightarrow$ Behaviour at $r \rightarrow \infty$

Ignore V_{eff} :
$$-\frac{\hbar^2}{2m} \frac{d^2 U_{El}}{dr^2} = E U_{El}(r)$$

For $E < 0$:
$$U_{El} \sim \exp\left(-\sqrt{\frac{2m|E|}{\hbar^2}} r\right)$$

For $E > 0$:
$$U_{El} \sim \exp(\pm ikr) \quad , \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

TODO: Generate discrete spectrum.

